

Supplementary Material

Comparative Modeling of Glucose Dynamics in Patients with Type 1 Diabetes based on Simulated and Real Data

Section S1: Clinical Safety Assessment via Clarke Error Grid Analysis (CEGA)

To rigorously evaluate the clinical safety of blood glucose forecasts beyond traditional statistical error metrics, predictions were evaluated using Clarke Error Grid Analysis (CEGA). CEGA maps predicted glucose concentrations against reference measurements across five distinct risk zones. Zone A represents predictions that fall within clinically accurate boundaries, leading to correct treatment decisions. Zone B encompasses predictions that deviate from reference values but remain clinically benign, causing no inappropriate treatment actions. Zones C, D, and E represent increasingly severe prediction errors that could result in unnecessary or harmful clinical interventions. Below are the visual grid distributions across short-, medium-, and long-term prediction horizons for Linear Regression (LR), Long Short-Term Memory (LSTM) networks, and eXtreme Gradient Boosting (XGBoost) models.

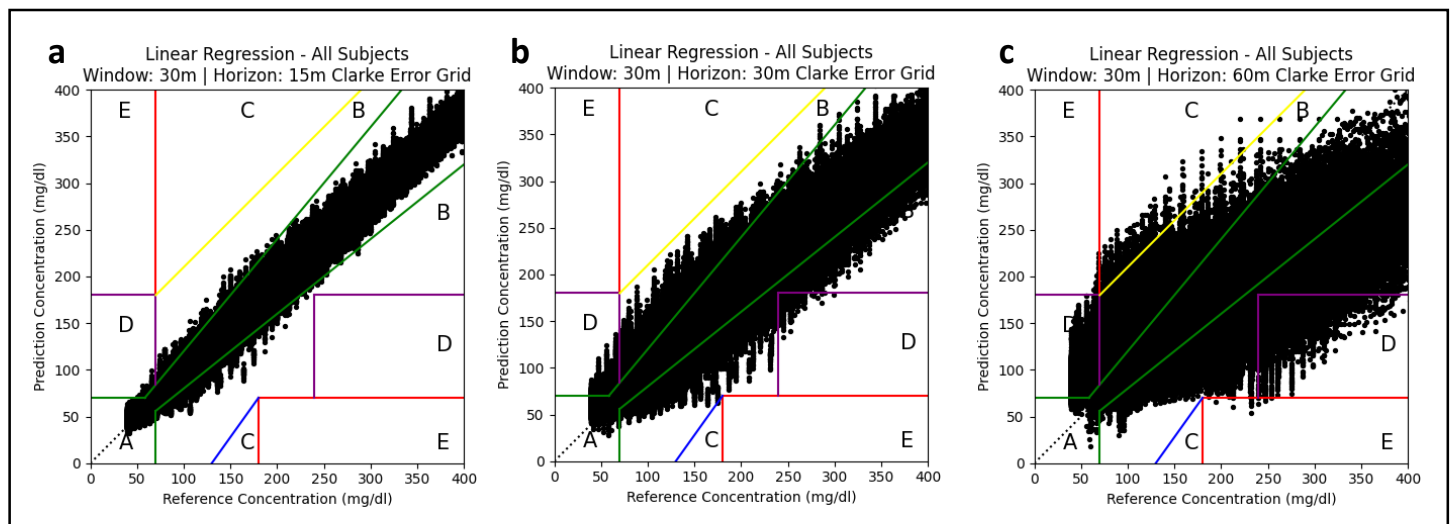


Figure S1.1 Performance of the linear regression model trained on the AZT1D dataset and evaluated on the Shanghai dataset using 30 minutes of historical data as input. The model performance is presented for three different prediction horizons: (a) 15 minutes, (b) 30 minutes, and (c) 60 minutes.

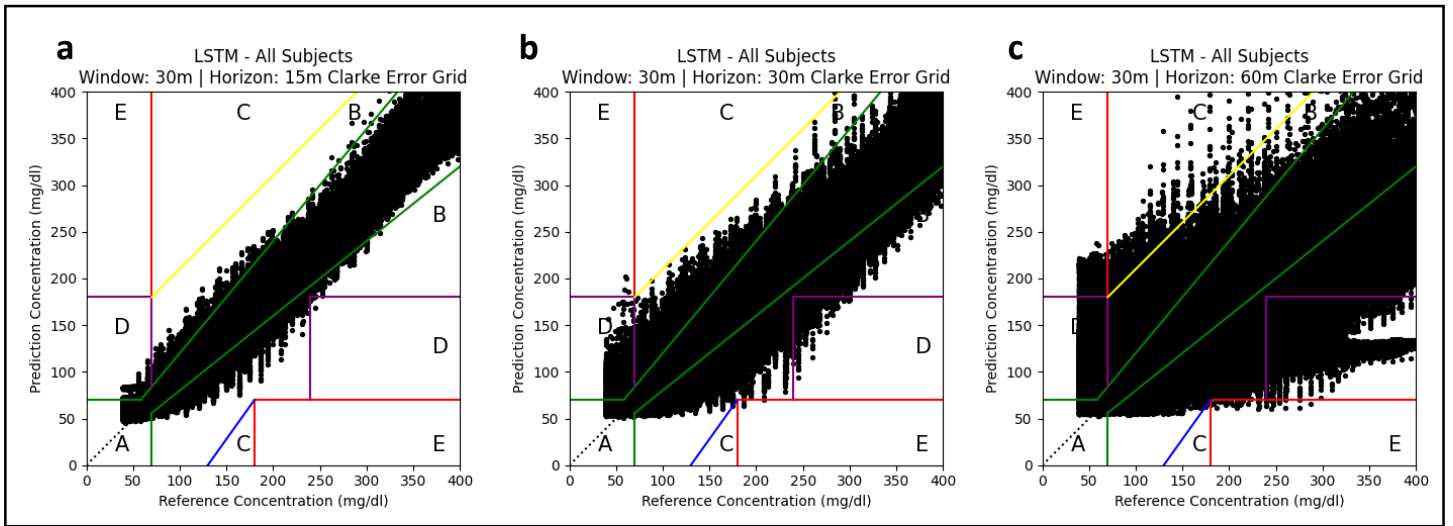


Figure S1.2 Performance of the LSTM model trained on the AZT1D dataset and evaluated on the Shanghai dataset using 30 minutes of historical data as input. The model performance is presented for three different prediction horizons: (a) 15 minutes, (b) 30 minutes, and (c) 60 minutes.

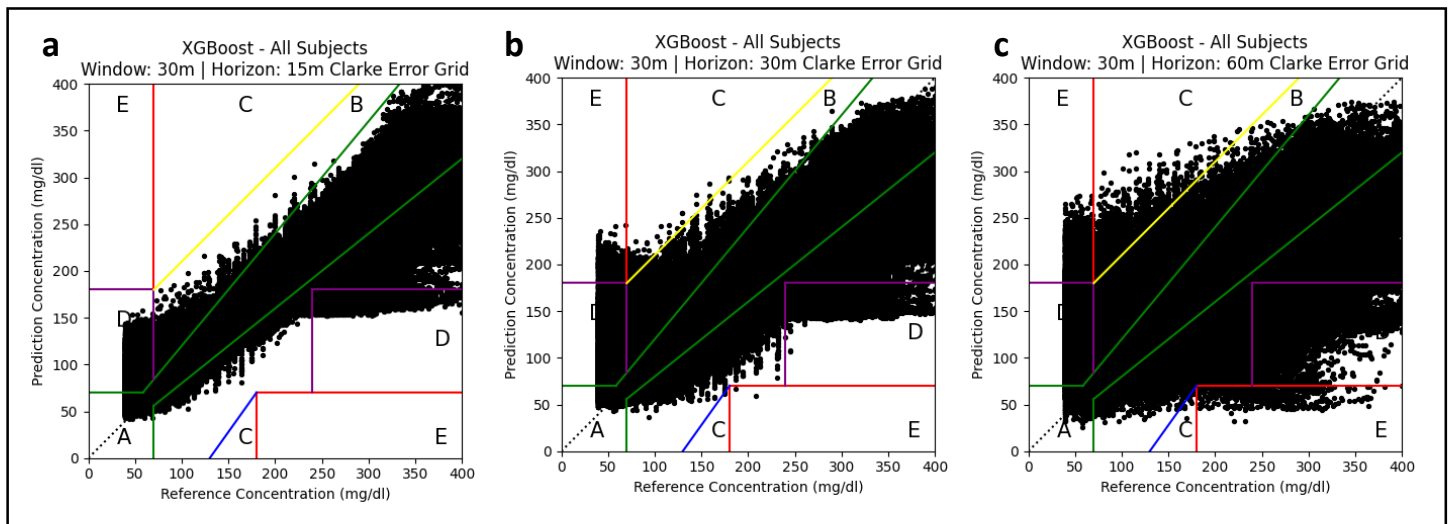


Figure S1.3 Performance of the XGBoost model trained on the AZT1D dataset and evaluated on the Shanghai dataset using 30 minutes of historical data as input. The model performance is presented for three different prediction horizons: (a) 15 minutes, (b) 30 minutes, and (c) 60 minutes.

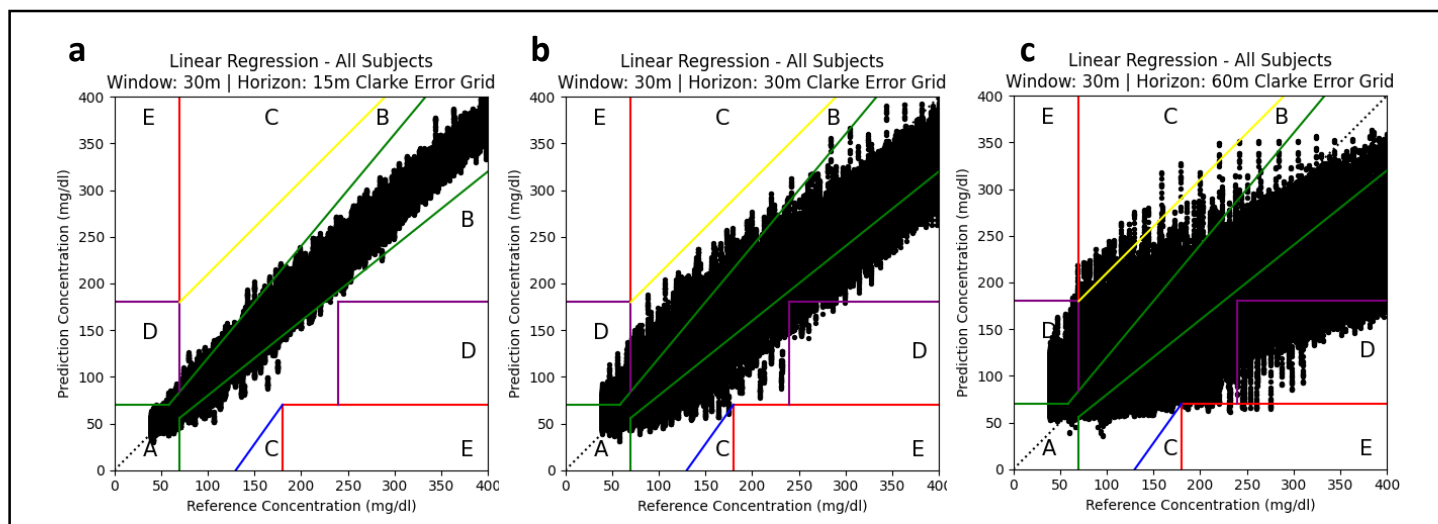


Figure S1.4 Performance of the linear regression model trained on simulated data from UVa-Padova simulator and evaluated on the Shanghai dataset using 30 minutes of historical data as input. The model performance is presented for three different prediction horizons: (a) 15 minutes, (b) 30 minutes, and (c) 60 minutes.

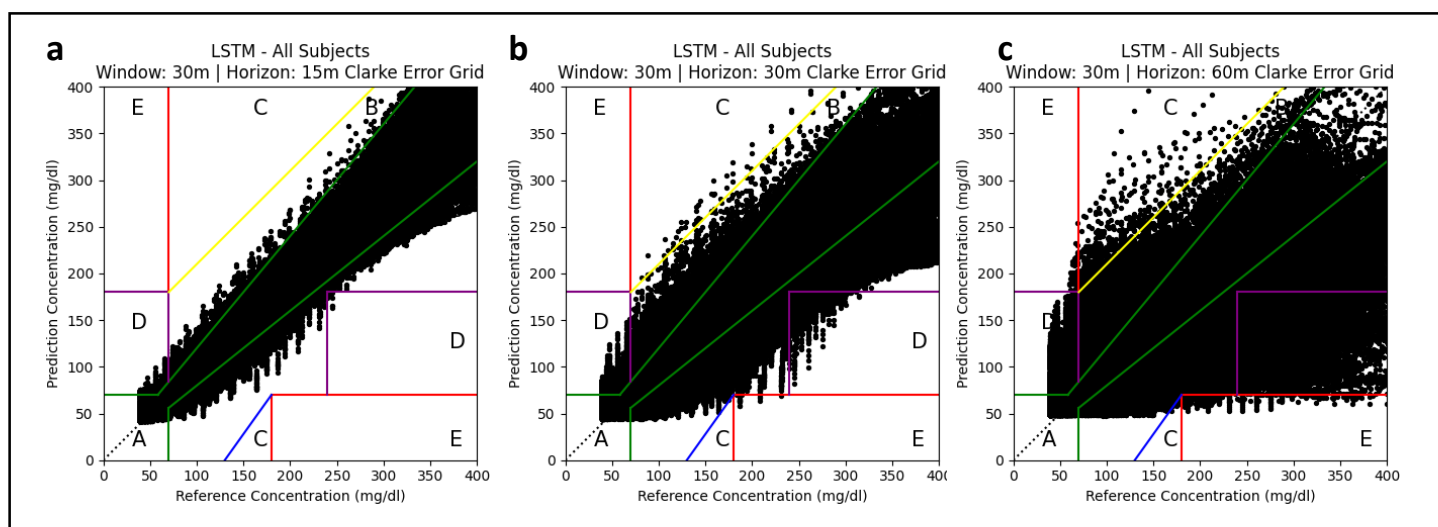


Figure S1.5 Performance of the LSTM model trained on simulated data from UVa-Padova simulator and evaluated on the Shanghai dataset using 30 minutes of historical data as input. The model performance is presented for three different prediction horizons: (a) 15 minutes, (b) 30 minutes, and (c) 60 minutes.

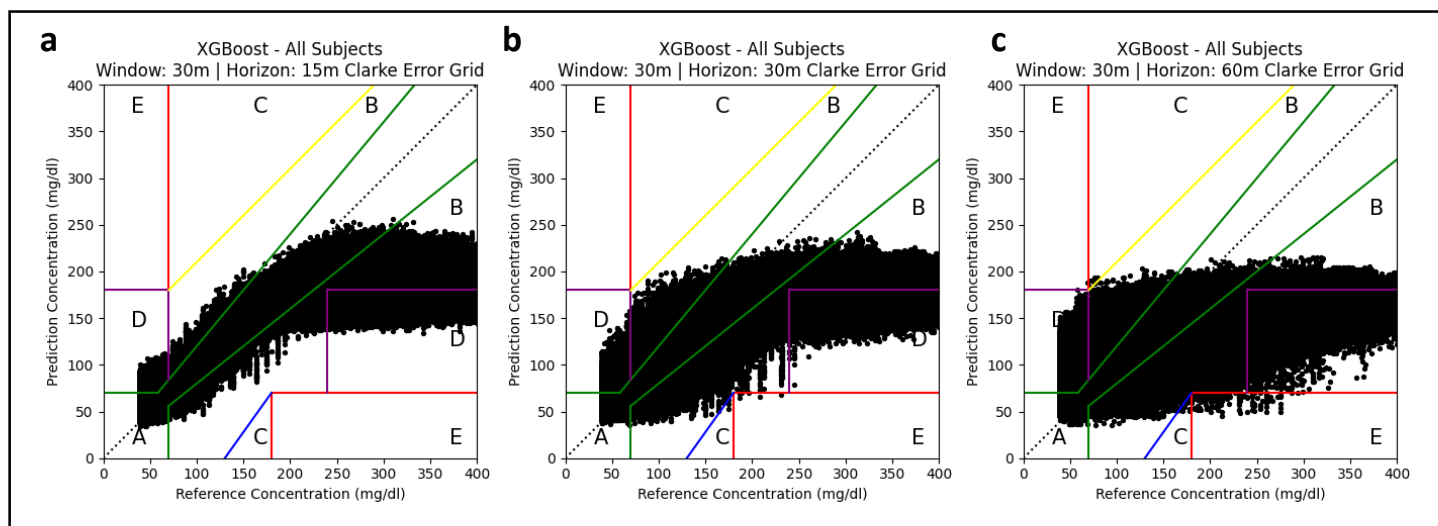


Figure S1.6 Performance of the XGBoost model trained on simulated data from UVA-Padova simulator and evaluated on the Shanghai dataset using 30 minutes of historical data as input. The model performance is presented for three different prediction horizons: (a) 15 minutes, (b) 30 minutes, and (c) 60 minutes.

Section S2: Cross-Cohort Model Fit Stability and RMSE Score Distributions

Evaluating model generalizability requires testing performance across heterogeneous patient populations and synthetic environments. The Root Mean Squared Error (RMSE) assesses how effectively each algorithm captures blood glucose dynamics when transferred between different cohorts, including the AZT1D clinical dataset, the ShanghaiT1DM clinical dataset, and the UVA-Padova simulator. The following box plots summarize the distribution, median performance, and interquartile variance of RMSE scores across models and forecasting horizons under four distinct prediction scenarios.

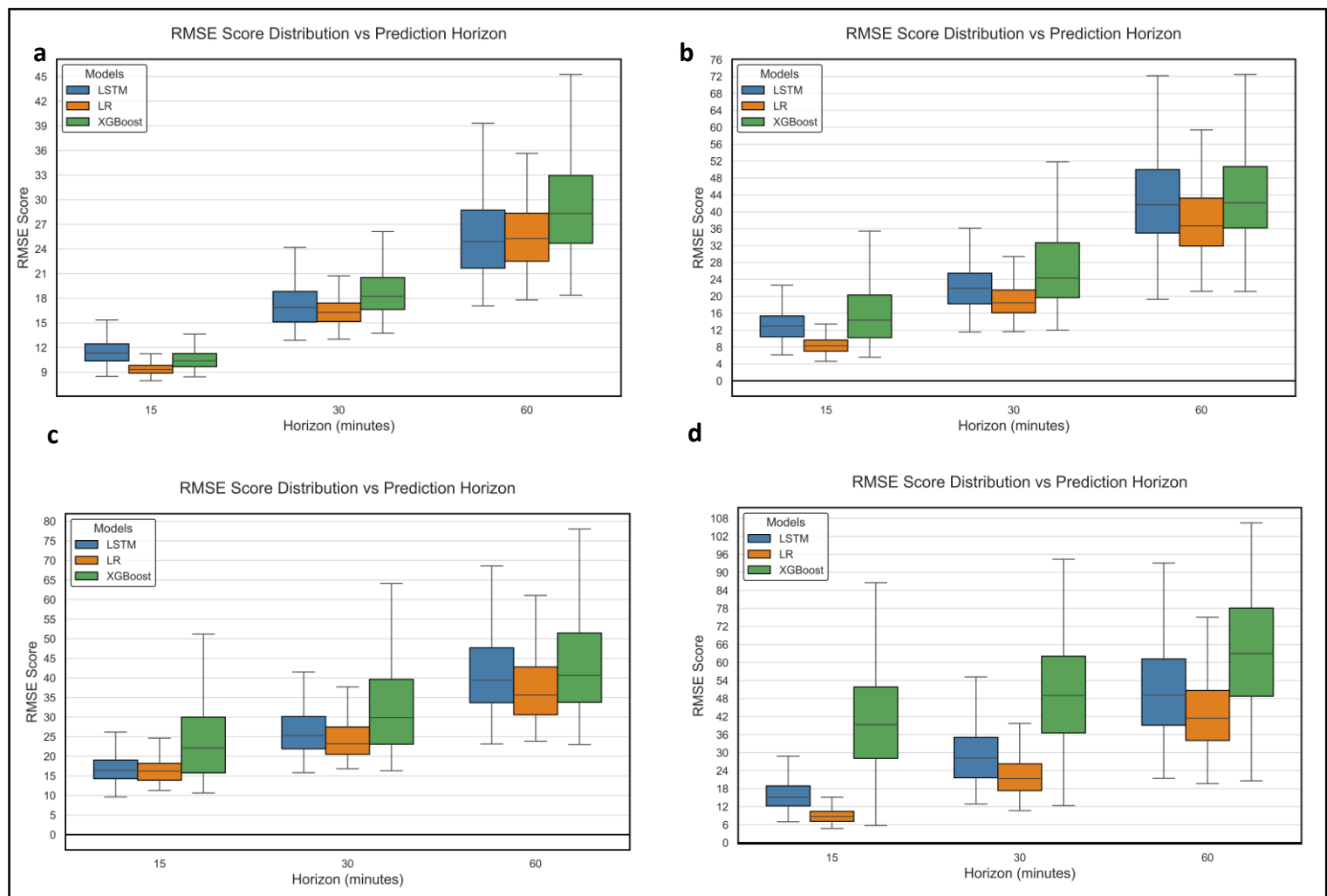


Figure S2.1 Forecasting performance of LSTM, LR, and XGBoost models across 15-, 30-, and 60-min prediction horizons, evaluated using RMSE. (a) Models trained on simulated UVA/Padova data and evaluated on the AZT1D dataset. (b) Models trained on simulated UVA/Padova data and evaluated on the Shanghai dataset. (c) Models trained on the AZT1D dataset and evaluated on simulated UVA/Padova data. (d) Models trained on the AZT1D dataset and evaluated on the Shanghai dataset.

Section S3: Historical Input Data Window Duration Sensitivity

To investigate whether providing models with longer historical glucose context improves forecasting accuracy, sensitivity analyses were conducted comparing a 30-minute historical input window against a 60-minute historical input window. Root Mean Squared Error (RMSE) was measured across all prediction horizons for each model architecture. The visualizations below present the error rate comparisons across input window sizes for each cross-cohort experimental configuration.

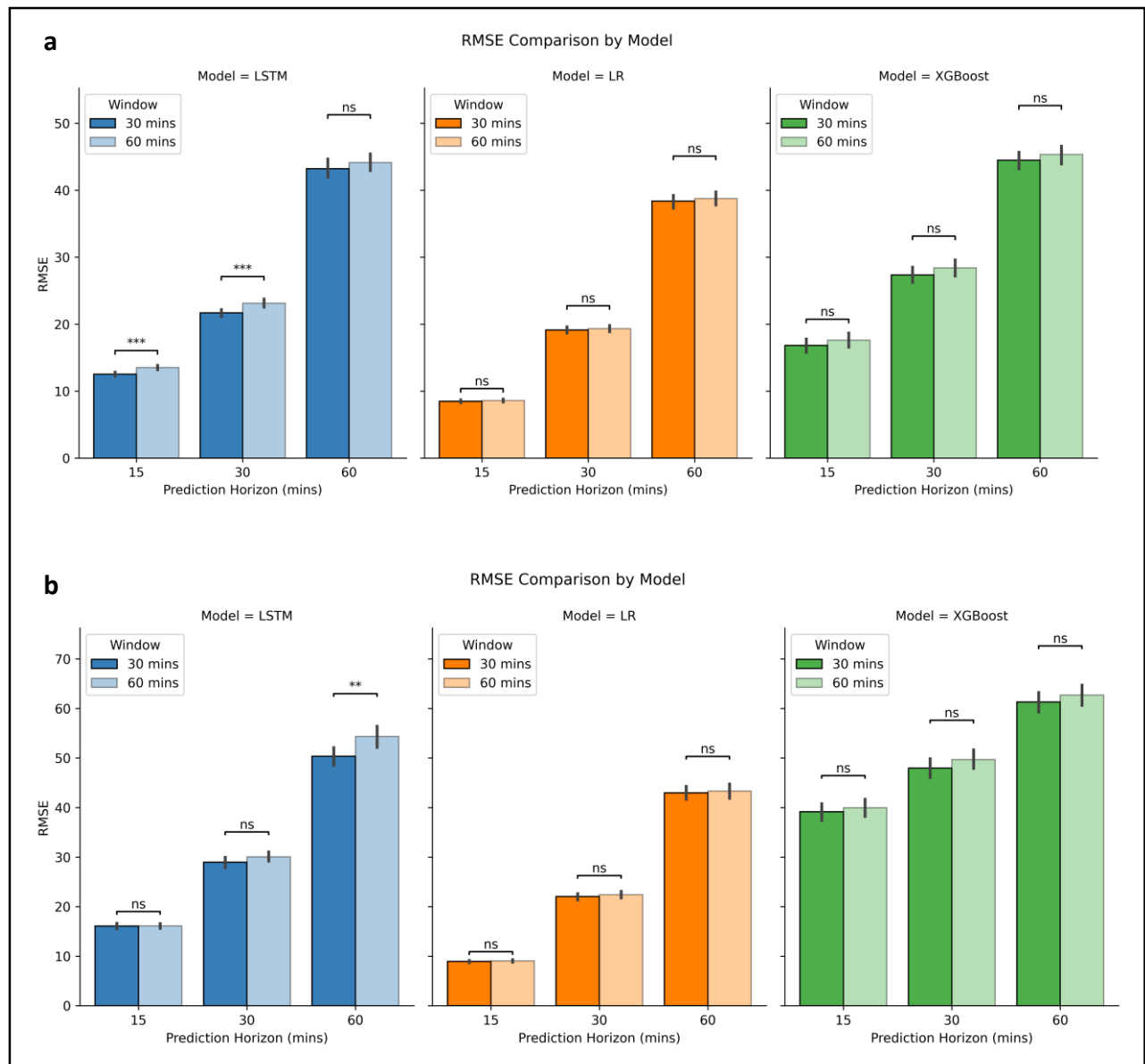


Figure S3.1 Comparing the impact of the past data window size in forecasting performance of three models – LSTM, LR and XGBoost based on RMSE performance. (a) Models trained the AZT1D dataset and evaluated on the Shanghai dataset. (b) Models trained on simulated UVA/Padova data and evaluated on the Shanghai dataset. (ns: non-significant, *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$ based on Welch's test)